

Efficient and Scalable Search for Statistics

42nd IEEE International Conference on Data Engineering (ICDE)

Antoine Gauquier¹ Simon Ebel²
Helena Galhardas³ Théo Galizzi² Ioana Manolescu²
Aurélien Peden² Pierre Senellart¹

¹DI ENS, ENS, CNRS, PSL University & Inria, Paris, France

²Inria & Institut Polytechnique de Paris, Palaiseau, France

³INESC-ID & IST, Universidade Lisboa, Lisbon, Portugal

May 8, 2026



PSL



Inria



U LISBOA

UNIVERSIDADE
DE LISBOA

Table of Contents

- 1 Context
- 2 Statistical Tables & TD Methods
- 3 STAR: Space- and Time-aware Statistic Retrieval
- 4 Experiments

Statistics in the Public Debate

Public debates rely on **metrics**: Inflation, growth, (un)employment, education, life expectancy, pollution, etc.

The value of a **metric**, on a given point of **time**, and for a given **spatial scale**, is a useful (and widely used) ingredient.

These **metrics**, generally called **statistics**, are produced by:

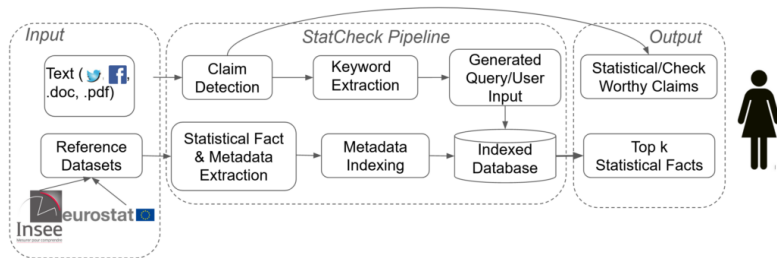
- *National agencies*: INSEE (France), Labor/Census Bureaus (US), ABS (AU)
- *International bodies*: Eurostat, IMF, WB, UN Agencies
- *Smaller entities*: Trade unions, NGOs



Statistics in the Public Debate

Access to **statistical data** is crucial:

- To **disseminate reliable information** to the broad public
- To **support journalists**, enabling efficient work in an environment flooded with continuous information
- To **counter misinformation**, especially on social media → enables (semi-)automatic **fact-checking**



Challenges in Accessing Statistical Data

Statistics are **publicly available**, but **finding metric values is hard**:

- *Heterogeneous formats*: CSV, Excel, HTML tables, JSON, SDMX (XML), etc.
 - *Sometimes compressed*, potentially with other content
 - Tables with up to *millions of rows* (e.g., in Eurostat)
- 1 We need automatic ways to **query large corpora of statistical tables**
 - 2 The targeted audience are broad public/journalists → **queries must be in natural language**

Table of Contents

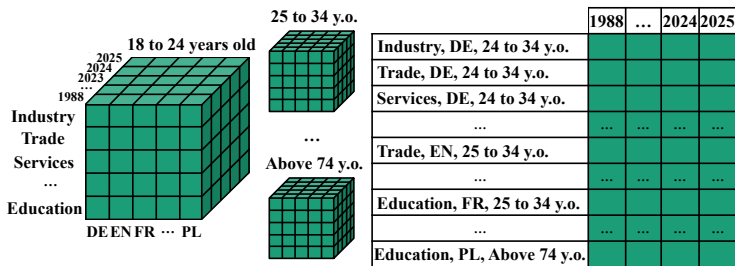
- 1 Context
- 2 Statistical Tables & TD Methods
- 3 STAR: Space- and Time-aware Statistic Retrieval
- 4 Experiments

Statistical vs Relational Tables

Statistical tables represent **multidimensional data**:

- E.g., “New jobs created” by year, country, sector, age group: each statistical value (*metric*) is defined for a set of d *dimension values*.
- Flattened into 2D tables for compatibility with classical tools: **serialization**.

Number of new jobs created across the European Union



Statistical vs Relational Tables

Challenges of serialized statistical tables:

- Key semantics are in row or column headers; cells are mostly numerical (no semantic cues).
- High dimensionality leads to very large tables (one row per combination of dimension values).
- Context for dimension values often missing.

Can traditional systems (for relational tables) handle this complexity?

Table Discovery Methods

Table Discovery (TD) → Finding **relevant tables** within a corpus in response to **natural language queries**.

Recent methods [WF23, BAW⁺25, GHM⁺25] leverage DL and LLMs:

- Annotated data generation
- Table schema synthesis
- Result re-ranking

They are designed for **relational tables** with well-defined schemas.

Two core challenges:

- 1 Can they efficiently address the **Statistical Table Discovery (STD)** problem, i.e., the TD problem over statistical tables?
- 2 Is it possible to come up with a **more frugal method** to be used by a broader public (journalists, citizens)?

Table of Contents

- 1 Context
- 2 Statistical Tables & TD Methods
- 3 STAR: Space- and Time-aware Statistic Retrieval**
- 4 Experiments

The STAR Approach

STAR = Space- and Time-aware **Statistic** Retrieval

At **indexing time**, STAR:

- ❶ Ignores numerical values → indexes only **linguistic elements** (titles, row and column headers)
 - ❷ **Spatio-temporal entities** are **extracted**
 - ❸ Aggregates linguistic elements and **projects them in a latent space** via a LM → store vectors in a multidimensional index
 - ❹ **Extracted spatio-temporal** elements are **indexed separately** using standardized hierarchies:
 - A Directed-Acyclic Graph for space (through reference sources, e.g., GeoNames)
 - A “natural” time hierarchy (through regular expressions)
- STAR **reasons about spatio-temporal covering**

The STAR Approach

At **query time**, STAR:

- ① Takes as input **an NL question**
- ② Question is parsed to **extract spatio-temporal entities**
- ③ Computes the subset of tables matching the constraints
- ④ Tables of the subset are ranked by **semantic similarity** with the embedding of the question
- ⑤ If no tables are found, the spatio-temporal constraints are **progressively relaxed** until having $k \geq 1$ matches.

Table of Contents

- 1 Context
- 2 Statistical Tables & TD Methods
- 3 STAR: Space- and Time-aware Statistic Retrieval
- 4 Experiments**

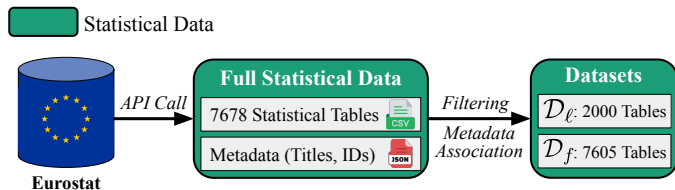
Experiments: Datasets

Dataset: $\sim 7\,600$ Eurostat tables in CSV format ($\mathcal{D}_f$):

- As **serialized tables**
- With $d \geq 1$ **dimensions**: always includes time, and in most cases space
- Along with their **titles** (and IDs)

Subset of 2 000 smallest tables used for scalability tests ($\mathcal{D}_\ell$).

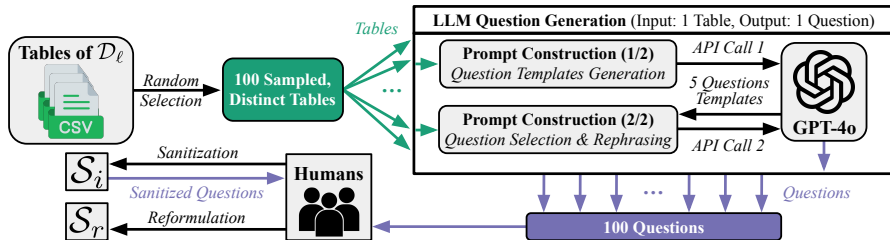
Dataset	#Tables	Size (MB)	#rows	#columns
$\mathcal{D}_\ell$	2 000	30.3	111 ($\pm$ 101)	19 ($\pm$ 23)
$\mathcal{D}_f$	7 605	98 521.7	69 077 ($\pm$ 318 706)	41 ($\pm$ 270)



Experiments: Question Sets

Two **questions sets** (for evaluation):

- **Initial** ($\mathcal{S}_i$): Lexically close to the table content
- **Reformulated** ($\mathcal{S}_r$): Same semantic meaning as $\mathcal{S}_i$, but with maximal lexical variation



Results in two **high-quality, manually sanitized** question sets
Evaluating on both **lexical** and **semantic** understanding

We score each system with 2 metrics:

- **HitRate@k**: Proportion of questions from a question set for which a system returned the *golden* table in its top- k ($1 \leq k \leq 10$).
→ **Standard metric in the literature**
- **Relevance@k**: measures a system's ability to return semantically relevant tables in its top- k results ($1 \leq k \leq 5$).
→ **More interesting, as there might be ≥ 1 relevant tables**

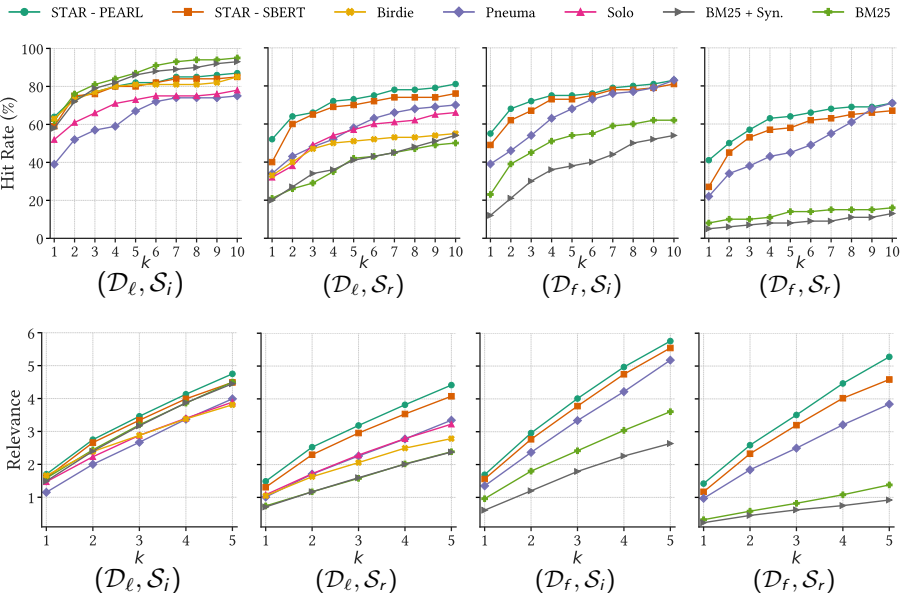
Experiments: Results – Running Time

Running time of the systems, for [indexing](#) and [querying](#):

Model	Indexing Time (s)			Query Time (s)		
	$\mathcal{D}_\ell$	$\mathcal{D}_f$	$(\mathcal{D}_\ell, \mathcal{S}_i)$	$(\mathcal{D}_\ell, \mathcal{S}_r)$	$(\mathcal{D}_f, \mathcal{S}_i)$	$(\mathcal{D}_f, \mathcal{S}_r)$
BM25	13	7 701	0.95	0.91	41.83	45.23
BM25+Syn.	13	7 701	3.39	3.34	47.16	51.47
Solo	81 847	OOM	1 267.05	1 143.06	OOM	OOM
Pneuma	45 300	532 020	3 793.23	3 680.90	1 770.44	1 814.24
Birdie	24 135	OOM	20.83	20.32	OOM	OOM
STAR - SBERT	371	45 103	36.77	37.66	186.62	190.95
STAR - PEARL	208	44 279	29.68	34.83	106.19	116.68

- Lexical-only methods scale and are the fastest
- [Solo](#) and [Birdie](#) do not scale to large datasets (OOM)
- For indexing: STAR is $12\times$ to $393\times$ faster than TD systems
- For querying: Slower than Birdie on $\mathcal{D}_\ell$, but $16\times$ faster than the only scaling TD system (Pneuma) on $\mathcal{D}_f$

Experiments: Results – Metrics



Takeaways

- ① We formalize the **STD** problem, and the challenges of **statistical data**
- ② We present **STAR**, a **space** and **time**-aware statistic retriever, operating over query semantics
- ③ We provide a **large novel benchmark** for the STD task:
 - 7 600 **statistic tables** from Eurostat (98 GB),
 - 200 **carefully curated questions** for lexical and semantic evaluation,
 - nearly 5 000 **manually annotated** (question, table) pairs
- ④ **STAR outperforms prior TD systems** and **lexical-only methods**:
 - Some competing systems **fail to scale** (OOM errors)
 - Even when close in score → STAR is at least **12× faster**
 - STAR requires **no GPU** (unlike LLM-based TD systems)

Thank you for your attention!
Any questions?

STAR code

`https://gitlab.inria.fr/cedar/star-statcheck`

Benchmark & extended version

`https://github.com/AntoineGauquier/efficient\_and\_scalable\_search\_for\_statistics/`

Bibliography I



Muhammad Imam Luthfi Balaka, David Alexander, Qiming Wang, Yue Gong, Adila Krisnadhi, and Raul Castro Fernandez.

Pneuma: Leveraging LLMs for Tabular Data Representation and Retrieval.
Proc. ACM Manag. Data, 3(3):200:1–200:28, 2025.



Lihu Chen, Gael Varoquaux, and Fabian Suchanek.

Learning High-Quality and General-Purpose Phrase Representations.

In Yvette Graham and Matthew Purver, editors, *Findings of the Association for Computational Linguistics: EACL 2024*, pages 983–994, St. Julian's, Malta, 2024. Association for Computational Linguistics.



Yuxiang Guo, Zhonghao Hu, Yuren Mao, Baihua Zheng, Yunjun Gao, and Mingwei Zhou.

Birdie: Natural Language-Driven Table Discovery Using Differentiable Search Index.

ArXiv preprint, abs/2504.21282, 2025.



Nils Reimers and Iryna Gurevych.

Sentence-BERT: Sentence embeddings using Siamese BERT-networks.

In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China, 2019. Association for Computational Linguistics.



Qiming Wang and Raul Castro Fernandez.

Solo: Data Discovery Using Natural Language Questions Via A Self-Supervised Approach.

Proc. ACM Manag. Data, 1(4):262:1–262:27, 2023.